

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
trades = pd.read_csv("historical_data.csv")
sentiment = pd.read_csv("fear_greed_index.csv")

print(trades.shape)
print(sentiment.shape)
```

```
(88467, 16)
(2644, 4)
/tmp/ipykernel_2955/3526668914.py:1: DtypeWarning: Columns (12) have mixed types
trades = pd.read_csv("historical_data.csv")
```

```
trades['Timestamp IST'] = pd.to_datetime(trades['Timestamp IST'])
trades['date'] = trades['Timestamp IST'].dt.date

sentiment['date'] = pd.to_datetime(sentiment['date']).dt.date
```

```
print(trades.columns)
print(sentiment.columns)
```

```
Index(['Account', 'Coin', 'Execution Price', 'Size Tokens', 'Size USD', 'Side',
      'Timestamp IST', 'Start Position', 'Direction', 'Closed PnL',
      'Transaction Hash', 'Order ID', 'Crossed', 'Fee', 'Trade ID',
      'Timestamp', 'date'],
      dtype='object')
Index(['timestamp', 'value', 'classification', 'date'], dtype='object')
```

```
merged = trades.merge(
    sentiment[['date', 'classification']],
    on='date',
    how='left'
)

print(merged.shape)
merged.head()
```

(88467, 18)

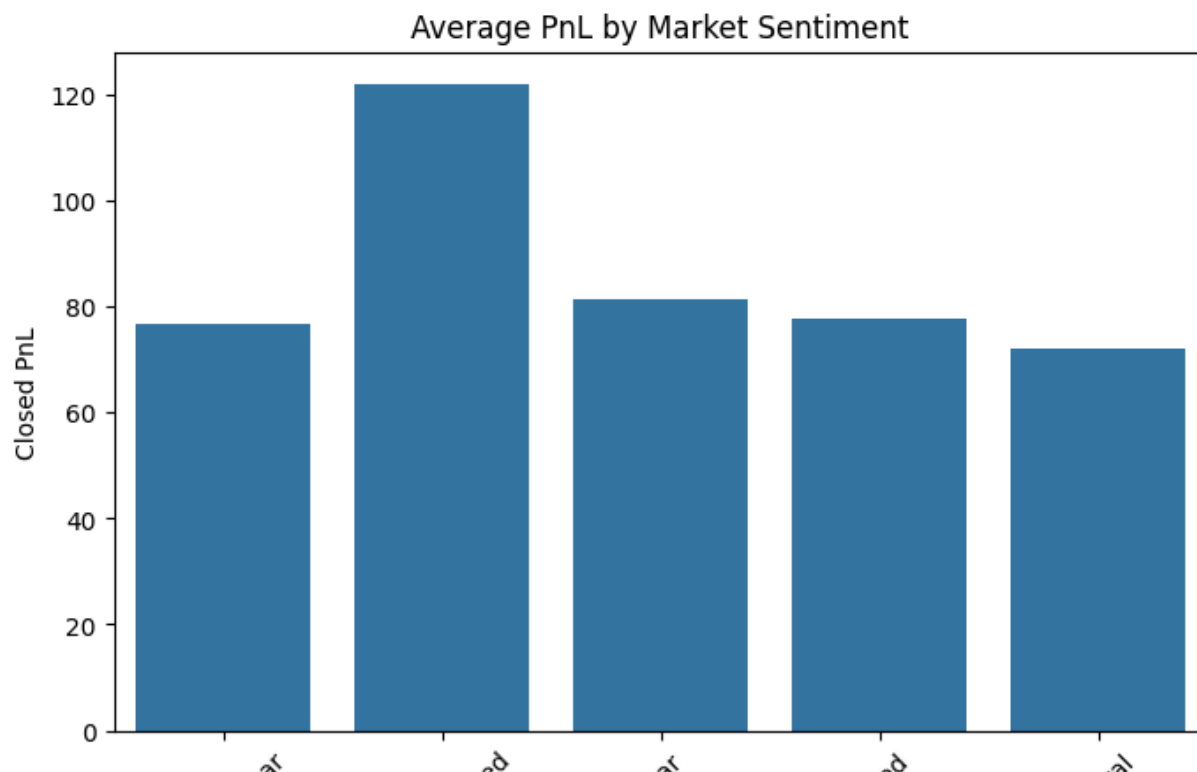
	Account	Coin	Execution Price	Size Tokens	Size USD
0	0xae5eacaf9c6b9111fd53034a602c192a04e082ed	@107	7.9769	986.87	7872.16
1	0xae5eacaf9c6b9111fd53034a602c192a04e082ed	@107	7.9800	16.00	127.68
2	0xae5eacaf9c6b9111fd53034a602c192a04e082ed	@107	7.9855	144.09	1150.63
3	0xae5eacaf9c6b9111fd53034a602c192a04e082ed	@107	7.9874	142.98	1142.04
4	0xae5eacaf9c6b9111fd53034a602c192a04e082ed	@107	7.9894	8.73	69.75

Next steps: [Generate code with merged](#) [New interactive sheet](#)

```
result = merged.groupby('classification')['Closed PnL'].mean()
print(result)
```

	classification	Closed PnL
0	Extreme Fear	76.680902
1	Extreme Greed	121.900517
2	Fear	81.463878
3	Greed	77.679216
4	Neutral	71.859323

```
plt.figure(figsize=(8,5))
sns.barplot(
    data=result,
    x='classification',
    y='Closed PnL'
)
plt.title('Average PnL by Market Sentiment')
plt.xticks(rotation=45)
plt.show()
```



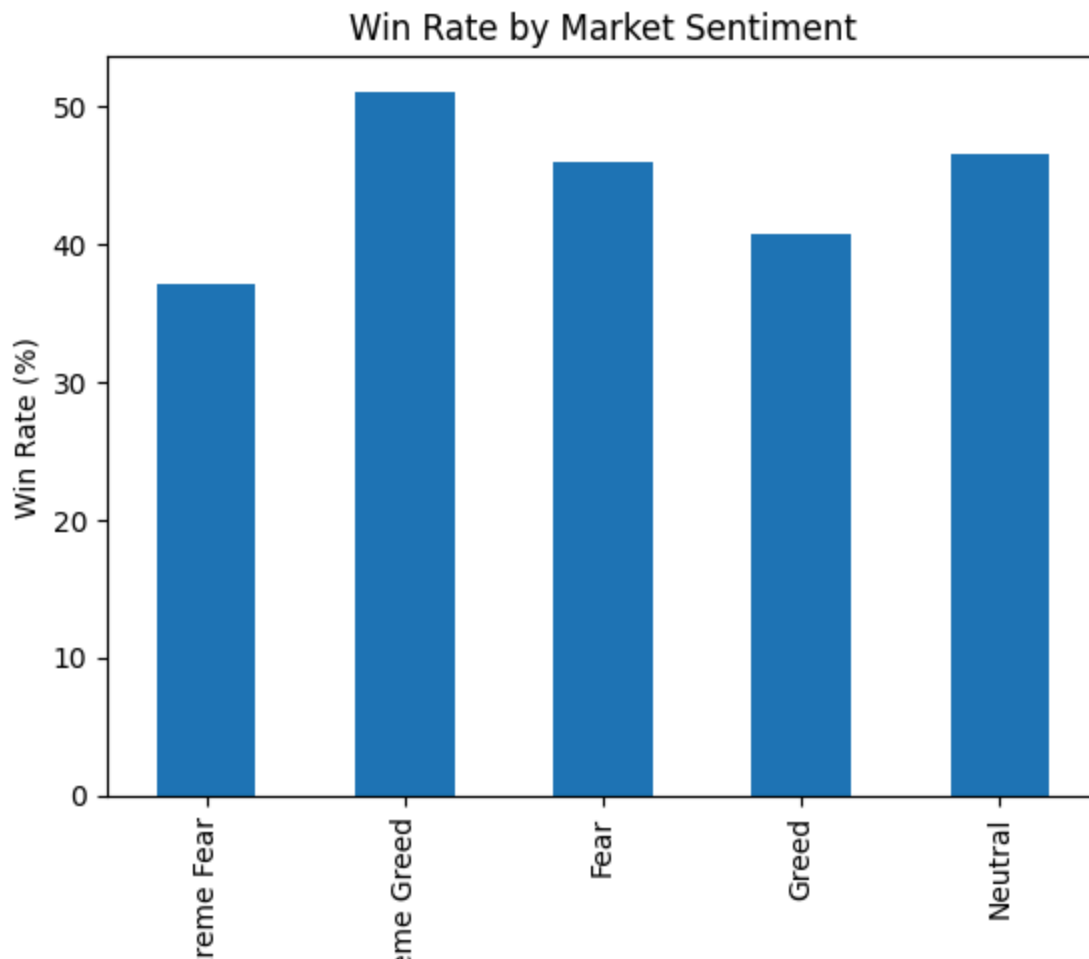
```
merged['win'] = merged['Closed PnL'] > 0

win_rate = merged.groupby('classification')['win'].mean()*100

print(win_rate)
```

```
classification
Extreme Fear      37.067922
Extreme Greed     51.058890
Fear              45.976449
Greed             40.724251
Neutral           46.565460
Name: win, dtype: float64
```

```
win_rate.plot(kind='bar')
plt.title('Win Rate by Market Sentiment')
plt.ylabel('Win Rate (%)')
plt.show()
```



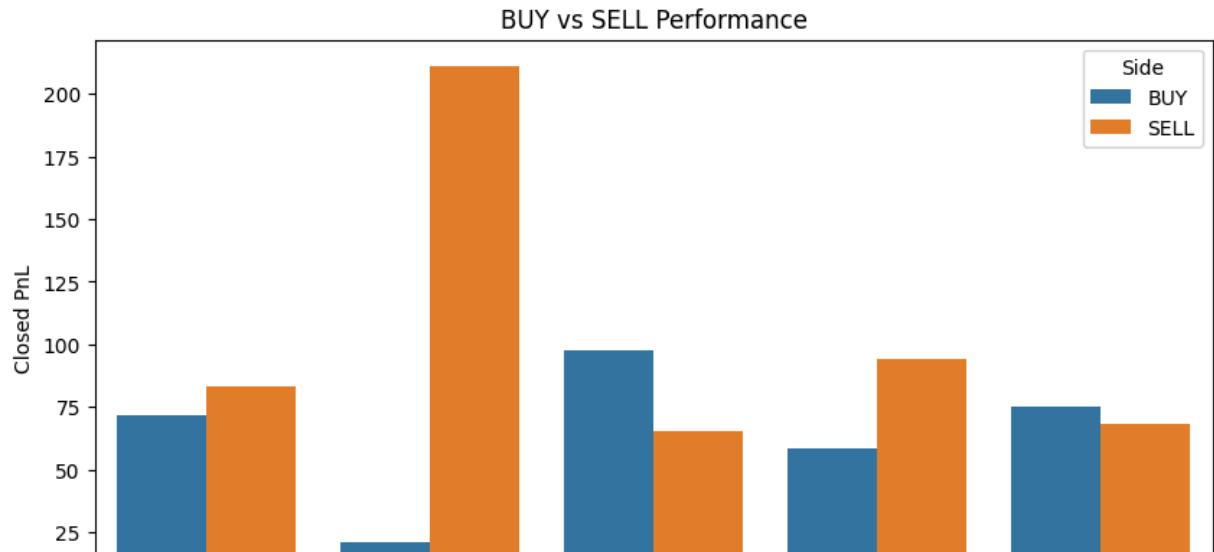
```
side_analysis = merged.groupby(
    ['classification', 'Side']
)['Closed PnL'].mean().reset_index()
```

```
print(side_analysis)
```

	classification	Side	Closed PnL
0	Extreme Fear	BUY	71.808992
1	Extreme Fear	SELL	83.142738
2	Extreme Greed	BUY	21.268160
3	Extreme Greed	SELL	210.986649
4	Fear	BUY	97.865214
5	Fear	SELL	65.519395
6	Greed	BUY	58.595557
7	Greed	SELL	94.393355
8	Neutral	BUY	75.254781
9	Neutral	SELL	68.268849

```
plt.figure(figsize=(10,5))
sns.barplot(
    data=side_analysis,
    x='classification',
    y='Closed PnL',
    hue='Side'
)
```

```
plt.title('BUY vs SELL Performance')
plt.show()
```



```
volume = merged.groupby(
    'classification'
)['Size USD'].sum().reset_index()

print(volume)
```

	classification	Size USD
0	Extreme Fear	3.915873e+07
1	Extreme Greed	8.473151e+07
2	Fear	3.310223e+08
3	Greed	2.264780e+08
4	Neutral	1.277166e+08

```
top_traders = merged.groupby(
    'Account'
)['Closed PnL'].sum().sort_values(
    ascending=False
).head(10)

print(top_traders)
```

Account	Closed PnL
0xb1231a4a2dd02f2276fa3c5e2a2f3436e6bfed23	2.143383e+06
0x083384f897ee0f19899168e3b1bec365f52a9012	1.600230e+06
0xbaaaf6571ab7d571043ff1e313a9609a10637864	9.401638e+05
0x513b8629fe877bb581bf244e326a047b249c4ff1	8.404226e+05
0x430f09841d65beb3f27765503d0f850b8bce7713	4.165419e+05
0x72c6a4624e1dffa724e6d00d64ceae698af892a0	4.030115e+05
0x75f7eeb85dc639d5e99c78f95393aa9a5f1170d4	3.790954e+05
0x4f93fead39b70a1824f981a54d4e55b278e9f760	3.089759e+05
0x420ab45e0bd8863569a5efbb9c05d91f40624641	1.995056e+05
0x6d6a4b953f202f8df5bed40692e7fd865318264a	1.087312e+05

Name: Closed PnL, dtype: float64

